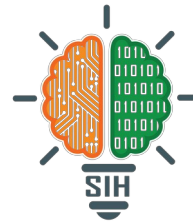


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2026

TITLE PAGE

Problem Statement ID – SIH26145

Problem Statement Title – AI-Based Detection of Cyber Threats in Unidirectional IP Traffic

Theme – Blockchain & Cybersecurity

PS Category – Software

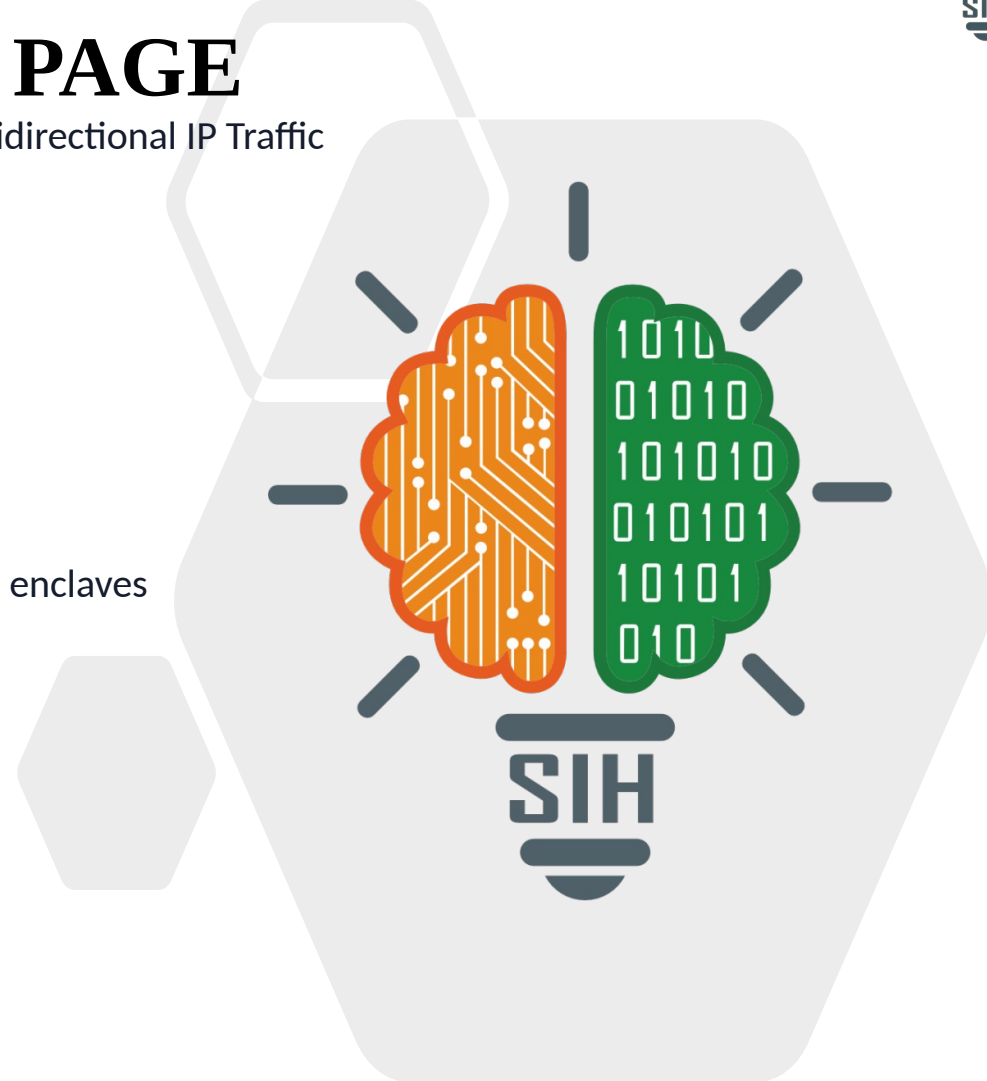
Organization – National Technical Research Organisation (NTRO)

Team ID – 145428

Team Name – ThreatNexus

Product: TRINETRA (त्रिनेत्र)

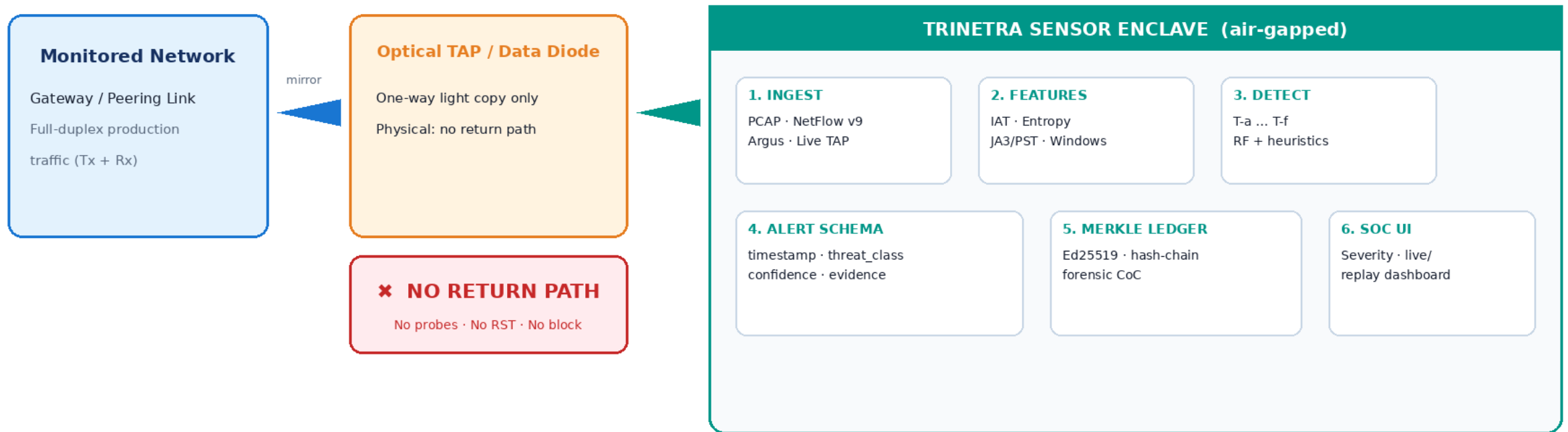
Passive AI threat detection for data-diode / unidirectional monitoring enclaves



IDEA TITLE

- ❖ TRINETRA — streaming AI/ML in a receive-only enclave: labelled alerts + confidence + evidence + forensic ledger (no return path, no decrypt, no inline block).

TRINETRA · Passive Diode-Enclave Threat Intelligence



Streaming (not batch)

Metadata only — zero decrypt

All 6 NTRO threat classes

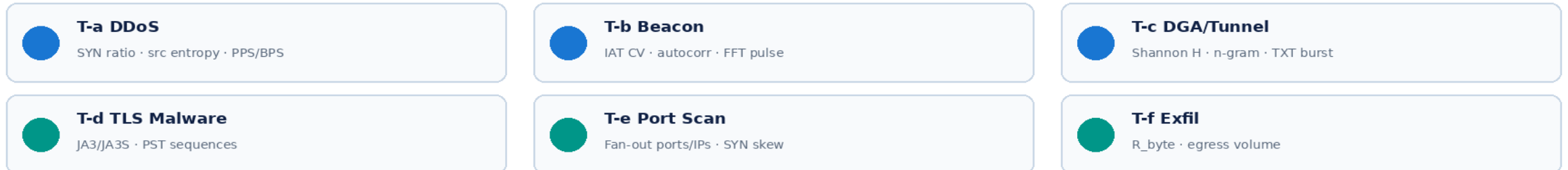
AST-verified zero-transmit

- Ingest → FlowTable → Windowed features → Detectors T-a...T-f → NTRO alert schema → Ed25519 Merkle ledger → SOC dashboard. Prototype in repo.

TECHNICAL PIPELINE · Streaming inference with bounded latency



Threat × Signal Matrix (PS a-f)



Stack (justified)

Python 3.11+ · dpkt PCAP · NetFlow v9 binary parser · scikit-learn RF · FastAPI SOC · Ed25519/cryptography · Rich TUI

Measured on live hardware (benchmark v5 · 50.2k packets)

~7.2k packets/sec	~21.5 Mbps sustained	~104 μs p50 service	~2.6 ms p50 alert lag	214 tests passed
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FEASIBILITY AND VIABILITY

- Feasible today as a working Python prototype: signed models, 214 tests, measured throughput — risks mapped to mitigations below.

FEASIBILITY · PS constraints → engineering mitigations

CHALLENGE / RISK	TRINETRA MITIGATION
No return path / no active NIDS	Passive TCP state + mid-stream recovery; AST/runtime no-transmit tests
Encrypted traffic (TLS/QUIC)	Metadata only: JA3/JA3S + PST — zero decryption (C2)
Alert flood / analyst fatigue	Incident-window deduplicator + severity + confidence
Air-gap: no cloud APIs	Local signed .joblib models; offline templates; Ollama optional/off
Demo realism & calibration	Synthetic PCAP gens + signed model cards + measured benchmarks
Theme: Blockchain without hype	Merkle + Ed25519 hash-chained ledger = forensic chain-of-custody

IMPACT AND BENEFITS

- Beneficiary: NTRO / critical-infra SOC analysts monitoring diode-isolated gateway & peering links.

IMPACT · NTRO / critical-infrastructure SOC enclaves

Operational

Earlier streaming alerts vs end-of-batch reports — without expanding the attack surface (monitor cannot pivot back).

Forensic

Tamper-evident Merkle ledger + Ed25519 enclave signatures → clean chain of custody for diode-isolated evidence.

National Security

Fits NTRO diode doctrine: passive intelligence on gateway/peering links for critical infrastructure.

Analyst UX

Labelled alerts with severity, confidence, and interpretable evidence features on a simple SOC dashboard.

Coverage

Single pipeline for all six PS threat classes (DDoS, C2, DGA/DNS tunnel, TLS malware, scan, exfil).

Future Path

Hardware diode pilot · live-traffic calibration · deeper ATT&CK mapping · JA4 upgrade path.

6/6

PS threat classes

214

automated tests

Read-only

ingest proven

Offline

air-gap ready

RESEARCH & REFERENCES · prior art + why diode changes the design

PS / Org

- SIH 2026 PS 26145 — NTRO unidirectional IP threat detection
- Data-diode / optical TAP monitoring doctrine (no return path)

Protocols & Fingerprints

- RFC 3954 NetFlow v9 · IPFIX · PCAP metadata ingest
- JA3/JA3S (Salesforce) · PST sequences · QUIC long-header metadata

Detection Science

- DGA: Shannon entropy + character n-gram language models
- C2 beaconing: IAT CV / autocorrelation / spectral periodicity
- Exfil: directional byte-ratio anomalies on passive flows

Threat Mapping

- MITRE ATT&CK: T1498, T1071, T1568.002, T1573, T1046, T1048
- Why Suricata-style active/inline NIDS is out of scope behind a diode

Our Artefacts

- Trinetra model cards · signed models_manifest.json · ps_compliance_matrix
- Measured benchmark_v5 JSON (throughput / latency / memory)